

Severe Combined Immunodeficiency Due To CARD11 Deficiency — Comprehensive Disease Report

MONDO ID: MONDO:0014081 · **Category:** Genetic (autosomal-recessive inborn error of immunity) · **Gene:** *CARD11* (HGNC:16393; 7p22.2) · **OMIM phenotype:** #615206 (Immunodeficiency 11A, IMD11A)

Summary

Severe combined immunodeficiency (SCID) due to CARD11 deficiency is an ultra-rare, autosomal-recessive inborn error of immunity caused by **biallelic loss-of-function (LOF) variants in *CARD11***, the lymphocyte-restricted scaffold protein that nucleates the **CARD11–BCL10–MALT1 (CBM) signalosome** downstream of the T- and B-cell antigen receptors. Complete absence of functional CARD11 abolishes assembly of the CBM complex and cripples antigen-receptor–driven activation of **NF-κB**, **c-Jun N-terminal kinase (JNK)**, and **MALT1 paracaspase**, together with a metabolic arm that couples the antigen receptor to **glutamine uptake and mTORC1**. The immunological hallmark is **agammaglobulinemia with profoundly deficient T-cell function despite numerically normal circulating T and B lymphocytes** — a functional SCID phenotype ([PMID: 23561803](#); [PMID: 33872653](#)).

Affected infants present early in life with life-threatening opportunistic infections (e.g., *Pneumocystis jirovecii* pneumonia, cytomegalovirus, recurrent viral and fungal respiratory infections), **interstitial lung disease**, and **severe colitis**. Because standard SCID newborn screening relies on quantifying **T-cell receptor excision circles (TRECs)**, which detect T-cell lymphopenia, CARD11 deficiency — with numerically normal T cells — can escape TREC-based screening, making functional immunology and genetic sequencing essential for diagnosis ([PMID: 42466165](#)). The definitive therapy is **allogeneic hematopoietic stem cell transplantation (HSCT)**, which is curative because CARD11 is lineage-restricted to hematopoietic tissue and the defect is intrinsic to lymphocytes ([PMID: 40625738](#); [PMID: 27109639](#)).

CARD11 sits at the center of a striking **allelic series**: biallelic null variants cause SCID/combined immunodeficiency; heterozygous gain-of-function (GOF) variants cause **BENTA** (B-cell Expansion with NF- κ B and T-cell Anergy); and heterozygous dominant-interfering LOF variants cause **severe atopic disease/CID**. *CARD11* deficiency is furthermore one of three interchangeable **CBM-complex combined immunodeficiencies** (with *BCL10* and *MALT1* deficiency) that phenocopy one another, defining the core differential-diagnosis group. This report synthesizes 10 confirmed findings from 42 reviewed papers across all requested disease-characteristic dimensions.

1. Disease Information

Overview. *CARD11*-deficiency SCID is a monogenic combined immunodeficiency in which complete loss of the *CARD11* scaffold prevents antigen-receptor signaling in T and B lymphocytes. It was first defined as a novel SCID entity by whole-exome sequencing in 2013 ([PMID: 23561803](#)). Clinically it behaves as a SCID/profound combined immunodeficiency (CID) but is biochemically distinct from lymphopenic SCIDs because lymphocyte *numbers* are preserved while lymphocyte *function* is lost.

Key identifiers.

Resource	Identifier
MONDO	MONDO:0014081
OMIM (phenotype)	#615206 — Immunodeficiency 11A (IMD11A)
Gene	<i>CARD11</i> , HGNC:16393, NCBI Gene 84433, UniProt Q9BXL7
Cytoband	7p22.2
Orphanet	Combined immunodeficiency due to <i>CARD11</i> deficiency (rare inborn error of immunity)
ICD-11	4A00.1 (combined immunodeficiencies)
MeSH	Severe Combined Immunodeficiency (D016511); CARD Signaling Adaptor Proteins

Synonyms / alternative names. Immunodeficiency 11A; CARD11 deficiency; complete CARD11 deficiency; combined immunodeficiency due to CARD11 deficiency; CARMA1 deficiency (CARD11 = CARMA1 = "CARD-containing MAGUK protein 1").

Source of information. Predominantly **aggregated disease-level and individual-patient case reports** — the literature comprises a small number of unrelated families/probands described in detail, plus mechanistic and structural studies. There is no large registry or EHR-derived cohort given the rarity.

2. Etiology

Disease causal factors. The disease is **purely genetic and monogenic**: biallelic (homozygous or compound-heterozygous) LOF variants in *CARD11* are necessary and sufficient. There is **no environmental, infectious, or lifestyle cause** — infections are consequences, not causes. Consanguinity is a recurrent contributing circumstance because it raises the chance of homozygosity for a rare recessive null allele (the index case was an infant of consanguineous parents; [PMID: 23561803](#)).

Genetic risk factors. The causal variants themselves are the risk factor. Reported pathogenic alleles include the nonsense variant **c.2509C>T**; **p.Arg837*** (recurrent, producing undetectable protein) and **p.Cys150*** ([PMID: 33872653](#); [PMID: 26289640](#)). No common susceptibility loci or modifier genes have been established for the recessive SCID form.

Environmental / protective factors. None established. The only "protective" genetic event documented is a **somatic second-site reversion** in a patient that partially restored CARD11 function and converted the phenotype toward Omenn syndrome ([PMID: 26289640](#)). In *hypomorphic* (not null) CARD11 disease, **glutamine supplementation** partially rescued downstream mTORC1/IFN- γ defects — a metabolic modifier relevant to residual-function alleles but not to complete deficiency ([PMID: 28628108](#)).

Gene–environment interactions. Not applicable in the classical GxE sense; the phenotype is fully genetically determined, though clinical severity is shaped by the pathogen exposures the immunodeficient host encounters.

3. Phenotypes

CARD11-deficiency SCID phenotypes fall into infection, immune-dysregulation/inflammatory, and laboratory categories. Onset is **neonatal to early-infancy**; severity is **severe**; course is **progressive/life-threatening without treatment**.

Phenotype	Type	Onset / severity	Suggested HPO term
Recurrent/opportunistic infections (<i>Pneumocystis jirovecii</i> pneumonia, CMV, sepsis)	Clinical sign	Neonatal–infancy; severe	HP:0002718 (Recurrent bacterial infections); HP:0002754 (Recurrent respiratory infections); HP:0032256 (Pneumocystis pneumonia)
Interstitial lung disease	Clinical/ imaging sign	Infancy; severe	HP:0006530 (Interstitial pulmonary abnormality)
Severe colitis / chronic diarrhea	Clinical sign	Infancy; severe	HP:0002583 (Colitis); HP:0002014 (Diarrhea)
Agammaglobulinemia / hypogammaglobulinemia	Laboratory abnormality	Congenital; severe	HP:0004432 (Decreased circulating total IgG); HP:0002090
Profoundly deficient T-cell function (poor proliferation to mitogens/antigens)	Laboratory abnormality	Congenital; severe	HP:0002843 (Abnormal T cell proliferation); HP:0005435 (Impaired T cell function)
Numerically normal T and B lymphocyte counts (distinguishing feature)	Laboratory finding	Congenital	(normal count — diagnostic caveat)
Omenn features on reversion (erythroderma, lymphoproliferation, elevated IgE, oligoclonal T cells)	Physical/ laboratory	Infancy; severe	HP:0001019 (Erythroderma); HP:0003212 (Increased circulating IgE); HP:0100827 (Lymphocytosis)
Failure to thrive	Physical manifestation	Infancy	HP:0001508 (Failure to thrive)

"presented with profound CID associated with viral and fungal respiratory infections, interstitial lung disease, and severe colitis" — [PMID: 33872653](#).

"The novel entity of SCID was characterized by agammaglobulinemia and profoundly deficient T-cell function despite quantitatively normal T and B lymphocytes." — [PMID: 23561803](#).

Frequency. With only a handful of reported patients, frequencies are qualitative: opportunistic infection, agammaglobulinemia, and defective T-cell function are essentially universal; interstitial lung disease and colitis are frequently reported; Omenn-syndrome presentation is exceptional (reversion-dependent).

Quality-of-life impact. Untreated disease is incompatible with normal life — recurrent life-threatening infection, chronic colitis with malabsorption, and respiratory compromise impose profound morbidity. Successful HSCT can restore near-normal immune function and quality of life.

4. Genetic / Molecular Information

Causal gene. *CARD11* (CARMA1), 7p22.2, encoding a 1154-aa multidomain membrane-associated guanylate kinase (MAGUK) scaffold with an N-terminal **CARD**, a **LATCH**, a **coiled-coil**, an autoinhibitory **inhibitory domain (ID)**, and C-terminal **PDZ-SH3-GUK** modules.

Pathogenic variants (recessive SCID form).

Variant (cDNA / protein)	Type	Consequence	Reference
c.2509C>T; p.Arg837*	Nonsense	Undetectable protein (complete deficiency); prevents CBM assembly	PMID: 33872653
Homozygous nonsense (truncating)	Nonsense	Truncated CARD11 defective in antigen-receptor signaling and NF-κB activation	PMID: 23561803
p.Cys150*	Nonsense	Impaired NF-κB signaling and IL-2 production	PMID: 26289640

Variant classification. Truncating null alleles are **pathogenic** (ACMG: PVS1 loss-of-function + segregation + functional data). Functional reconstitution assays confirmed patient-derived truncated CARD11 is signaling-defective ([PMID: 23561803](#)).

Variant type/class. Predominantly **nonsense/truncating LOF**. **Allele frequency:** these specific pathogenic nulls are extremely rare/private in gnomAD, consistent with a severe recessive disease. **Origin: germline** (the disease-causing alleles); note the exception of a

somatic second-site reversion that mitigated one patient's phenotype (PMID: 26289640). Somatic *CARD11* GOF mutations are separately associated with B-cell lymphomas (not this disease).

Functional consequence. Complete loss of function — abolition of the scaffold prevents any BCL10 nucleation. This contrasts with (a) **dominant-negative** heterozygous LOF alleles that poison wild-type *CARD11* (atopy/CID) and (b) **gain-of-function** alleles that bypass autoinhibition (BENTA / lymphoma).

The *CARD11* allelic series (a defining feature): "*Germline *CARD11* mutations cause several distinct primary immune disorders in human subjects, including severe combined immune deficiency (biallelic null mutations), B-cell expansion with nuclear factor κ B and T-cell anergy (heterozygous, gain-of-function mutations), and severe atopic disease (loss-of-function, heterozygous, dominant interfering mutations)*" — PMID: 30170123.

Modifier genes / epigenetics / chromosomal abnormalities. No established modifier genes for the recessive SCID form. No disease-specific epigenetic signature or chromosomal abnormality — this is a single-gene coding disorder.

5. Environmental Information

Not applicable as a cause. *CARD11*-deficiency SCID has **no environmental, lifestyle, or toxicological etiology**. Infectious agents (*Pneumocystis jirovecii*, cytomegalovirus, and other viral/fungal/bacterial pathogens) are **downstream consequences** of the immunodeficiency rather than triggers. Practical environmental relevance is limited to **infection-avoidance / protective isolation** and avoidance of **live vaccines** in the immunodeficient host prior to definitive therapy.

6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic null *CARD11* variant** (e.g., p.Arg837*) → **absent/non-functional *CARD11* scaffold protein** in lymphocytes (demonstrated: undetectable protein) — PMID: 33872653.

2. Absent CARD11 → **failure of receptor-triggered closed→open conformational activation** of the scaffold (CARD11 normally relieved from autoinhibition by PKC θ /β phosphorylation) — [PMID: 20799731](#).
3. No open CARD11 → **no nucleation of BCL10 filaments** and **failure of CARD11–BCL10–MALT1 (CBM) complex assembly** — [PMID: 31296852](#); [PMID: 33872653](#).
4. No CBM signalosome → **severely impaired activation of NF-κB, JNK, and MALT1 paracaspase** in both B and T cells (demonstrated) — [PMID: 33872653](#).
5. **Branch A (metabolic):** CBM loss → **impaired TCR-stimulated glutamine uptake and mTORC1 activation**, independent of IKK — [PMID: 24792914](#).
6. **Branch B (cytokine):** CBM loss → **impaired IL-2 production** — [PMID: 26289640](#).
7. Impaired NF-κB/JNK/MALT1 signaling → **developmental block in B cells at the naive/type-1 transitional stage** and **impaired circulating T follicular helper cells / defective T-cell effector function** — [PMID: 33872653](#).
8. Defective lymphocyte activation and differentiation → **agammaglobulinemia + profoundly deficient T-cell function despite normal lymphocyte counts** — [PMID: 23561803](#).
9. Loss of protective adaptive immunity → **early-life opportunistic infections, interstitial lung disease, colitis** → life-threatening combined immunodeficiency — [PMID: 33872653](#).

(Inferred branch: an Omenn-syndrome phenotype can arise when a somatic second-site reversion partially restores CARD11, generating oligoclonal autoreactive T cells — [PMID: 26289640](#).)

Detail by category

- **Molecular pathways.** Antigen-receptor (TCR/BCR) → PKC θ /β → **CARD11** → BCL10/MALT1 → IKK → **canonical NF-κB**; parallel arms to **JNK/AP-1** and **MALT1 protease**; and a metabolic arm to **glutamine/ASCT2 → mTORC1** ([PMID: 24792914](#); [PMID: 26260210](#)). Suggested pathway refs: KEGG NF-κB signaling; Reactome "CARD11-BCL10-MALT1 assembly."
- **Cellular processes.** Lymphocyte activation, proliferation, survival, and effector-lineage differentiation (Th17 differentiation requires CARMA1; [PMID: 23091043](#)); thymic Treg commitment (see model organisms).
- **Protein dysfunction.** Complete loss of scaffold function (null); by contrast, GOF variants disrupt the autoinhibitory ID interface causing spontaneous BCL10 filament formation — [PMID: 31296852](#); [PMID: 26884335](#).

- **Metabolic changes.** Loss of CARD11-dependent glutamine import → reduced mTORC1 activity → reduced IFN- γ (documented in hypomorphic disease, partially glutamine-rescuable) — [PMID: 28628108](#).
- **Immune-system involvement.** Combined (T + B) **immunodeficiency**; the immune failure is the disease.
- **Molecular profiling.** Reconstitution and reporter assays establish the NF- κ B defect; patient B-cell developmental staging documents the transitional block ([PMID: 33872653](#)).

Upstream vs downstream: the mutation and absent scaffold are upstream; NF- κ B/JNK/MALT1/mTORC1 failure is intermediate; lymphocyte developmental/functional defects and clinical infection are downstream.

Suggested GO / CL terms. GO:0007249 (I- κ B kinase/NF- κ B signaling); GO:0050852 (T-cell receptor signaling pathway); GO:0050853 (B-cell receptor signaling pathway); GO:0002250 (adaptive immune response). Cell types: CL:0000084 (T cell), CL:0000236 (B cell), CL:0002679 (regulatory T cell), CL:0000788 (naive B cell).

7. Anatomical Structures Affected

- **Organ / system level. Immune (lymphoid) system** primarily: thymus, bone marrow, spleen, lymph nodes, mucosa-associated lymphoid tissue. Secondary organ involvement from immune failure: **lungs** (interstitial lung disease; UBERON:0002048) and **gastrointestinal tract / colon** (colitis; UBERON:0001155). Systems involved: immune/hematopoietic; secondarily respiratory and digestive.
- **Tissue / cell level. Lymphocytes** — T cells (CL:0000084), B cells (CL:0000236), T follicular helper cells (CL:0002038), regulatory T cells (CL:0002679), naive/transitional B cells (CL:0000788). Hematopoietic tissue is the site of the intrinsic defect.
- **Subcellular level.** CARD11 acts at the **plasma membrane / cytoplasmic signalosome**; GO cellular components: GO:0032449 (CBM complex), GO:0005886 (plasma membrane), GO:0005829 (cytosol).
- **Localization / lateralization.** Systemic and **bilateral/diffuse** — a whole-body lymphoid disorder, not a focal lesion. The immune-restricted anatomy is explained by **lineage-restricted CARD11 expression**: "*CARD11 is primarily expressed in hematopoietic tissues and lymphocytes and plays a crucial role in the proper activation of B and T cells in response to antigen recognition*" — [PMID: 40625738](#).

8. Temporal Development

- **Onset. Congenital defect; clinical onset in the neonatal period/early infancy** with opportunistic infection — *"in patients with severe combined immunodeficiency (SCID), infections caused by opportunistic pathogens are typically life-threatening early in life"* — [PMID: 23561803](#). Onset pattern is **acute-on-chronic** (an underlying congenital immune defect punctuated by acute infections).
- **Progression. Progressive and life-threatening** without intervention; no spontaneous remission (except the rare partial, somatic-reversion event that alters rather than cures the phenotype). Disease duration is **lifelong** unless corrected by HSCT.
- **Critical period / window of opportunity.** Early diagnosis before overwhelming infection is critical; **HSCT before infectious/organ damage accrues** offers the best outcomes, as for other SCIDs.

9. Inheritance and Population

- **Inheritance. Autosomal recessive** (biallelic null). Consanguinity increases risk via homozygosity ([PMID: 23561803](#)). By contrast, dominant-negative LOF and GOF *CARD11* disorders are autosomal dominant — the same gene yields different inheritance depending on allele class ([PMID: 30170123](#); [PMID: 25645939](#)).
- **Penetrance / expressivity.** Biallelic null penetrance is presumed **complete** for the immunodeficiency; expressivity can vary (e.g., Omenn presentation via somatic reversion).
- **Epidemiology. Ultra-rare** — described in a small number of unrelated families worldwide; no reliable prevalence/incidence estimate exists. It is a small fraction of overall SCID incidence (SCID overall ~1 in 50,000–100,000 births).
- **Founder effects / carrier frequency.** No established founder allele; carrier frequency is expected to be very low, elevated within consanguineous kindreds.
- **Demographics.** No sex predilection (autosomal). Reported across different populations; consanguineous families are over-represented among cases.

10. Diagnostics

Laboratory / immunological. Serum immunoglobulins (**agammaglobulinemia/hypogammaglobulinemia**), **lymphocyte subset enumeration** (T and B counts **normal** — a key discriminator), and **functional T-cell proliferation** to mitogens (PHA) and antigens (**profoundly reduced**) ([PMID: 23561803](#)). Suggested LOINC-type panels: immunoglobulin quantitation, lymphocyte subset panel, lymphocyte mitogen stimulation.

Functional confirmation. **NF-κB activation and CBM assembly assays / reconstitution assays** demonstrating that patient-derived CARD11 is signaling-defective ([PMID: 23561803](#); [PMID: 33872653](#)).

Genetic testing. **WES/WGS or targeted PID gene panels including *CARD11*** are the definitive diagnostic modality; single-gene sequencing confirms biallelic variants and segregation. Chromosomal microarray/karyotype/FISH are not indicated (point mutations, not structural).

Newborn-screening caveat (critical). SCID NBS quantifies **TRECs**, which reflect **T-cell lymphopenia**; CARD11-deficient patients have **numerically normal T cells** and may therefore **screen normal (false-negative)**. *"Newborn screening (NBS) based on quantifying T cell receptor excision circles (TRECs) is highly sensitive for detecting severe combined immunodeficiency (SCID)"* — [PMID: 42466165](#) — but this sensitivity depends on lymphopenia the patient does not exhibit. **Second-tier genetic screening** can improve accuracy for such non-lymphopenic T-cell defects.

Differential diagnosis. Other **CBM-complex combined immunodeficiencies** — **BCL10** and **MALT1** deficiency — phenocopy CARD11 deficiency and constitute the primary differential group: *"a broad range of clinical manifestations, including those characteristic of T- and B-lymphocyte defects, are associated with CARD11, MALT1, and BCL10 deficiencies"* — [PMID: 26277595](#). **CARD9 deficiency** is distinguished because it causes **isolated invasive fungal infection** (innate CBM), not combined immunodeficiency ([PMID: 26277595](#)). Also consider other SCID/CID genes, hyper-IgE syndromes, and (for the reversion phenotype) Omenn syndrome.

11. Outcome / Prognosis

- **Natural history (untreated).** Poor — early-life opportunistic infection, interstitial lung disease, and colitis carry high mortality, consistent with SCID.

- **With treatment. Allogeneic HSCT is potentially curative.** By analogy within the CBM group, MALT1 deficiency has been **successfully treated and immunologically normalized by HSCT** (PMID: 27109639), and CARD11's lineage-restricted (hematopoietic) expression means donor immune reconstitution corrects the intrinsic defect (PMID: 40625738).
- **Morbidity / complications.** Chronic lung disease, colitis with malabsorption/failure to thrive, disseminated infection; transplant-related complications (GVHD, conditioning toxicity, viral reactivation).
- **Prognostic factors.** Age at diagnosis and transplant, organ/infectious damage at HSCT, donor match, and conditioning regimen. Early, pre-damage transplant favors better outcomes.

12. Treatment

Definitive therapy. - **Allogeneic hematopoietic stem cell transplantation (HSCT)** — the curative standard for CBM-complex combined immunodeficiencies, ideally with reduced-intensity conditioning; supported by successful normalization in MALT1 deficiency (PMID: 27109639). NCIT: C15431 (Hematopoietic Stem Cell Transplantation).

Supportive / bridging pharmacotherapy. - **Immunoglobulin replacement therapy (IVIG/ SCIG)** for agammaglobulinemia. NCIT: C29294 (Intravenous Immunoglobulin Therapy). - **Anti-infective prophylaxis** — *Pneumocystis jirovecii* prophylaxis (trimethoprim-sulfamethoxazole), antifungal and antiviral prophylaxis; treatment of intercurrent infections. - **Avoid live vaccines**; use irradiated/CMV-safe blood products.

Experimental / metabolic. In **hypomorphic** (residual-function) CARD11 disease, **glutamine supplementation** partially rescued mTORC1 and IFN- γ defects — a proof-of-concept metabolic intervention that requires residual CARD11 for glutamine import and is therefore **not expected to help complete null deficiency** (PMID: 28628108). Gene therapy/gene correction is conceptually attractive (hematopoietic-restricted, single-gene) but not clinically established for this disease.

Personalized-medicine note. Treatment is genotype-specific across the allelic series: null SCID → immune reconstitution (HSCT); GOF BENTA → immunosuppression (e.g., sirolimus/mTOR inhibition; PMID: 39998705). Correctly classifying the allele is essential to avoid opposite treatment errors.

13. Prevention

- **Primary prevention.** Not preventable at the individual level (germline recessive). Population-level: **genetic counseling and carrier testing** in consanguineous families or families with a prior affected child; **preimplantation/prenatal genetic diagnosis** and **cascade testing** of relatives once the familial variant is known.
 - **Secondary prevention.** Early detection — although **TREC NBS may miss CARD11 deficiency** (normal T-cell counts), **second-tier genetic screening** and a high index of suspicion in infants with hypogammaglobulinemia + poor T-cell function improve early diagnosis ([PMID: 42466165](#)).
 - **Tertiary prevention.** Infection prophylaxis, protective isolation, irradiated blood products, avoidance of live vaccines, and timely HSCT to prevent irreversible organ damage.
-

14. Other Species / Natural Disease

- **Taxonomy / orthologs.** Human *CARD11* has a conserved mouse ortholog ***Card11*** (a.k.a. *Carma1*; NCBI Gene 108723; *Mus musculus*, NCBI:txid10090). Orthologs exist across mammals; the CBM signaling module is evolutionarily conserved.
 - **Natural disease in other species.** No well-characterized naturally occurring companion-animal or wildlife disease is established for CARD11 in OMIA; disease knowledge derives from engineered mouse models rather than spontaneous animal disease.
 - **Comparative biology.** Mouse *Card11* loss recapitulates key lymphocyte-activation and Treg defects (below), but concordance with humans is partial: *"human subjects with these mutations have some features in common with the corresponding knockout mice, but other features are different between human subjects and mice"* — [PMID: 26277595](#).
 - **Zoonotic potential.** Not applicable (non-infectious genetic disease).
-

15. Model Organisms

- **Model type.** Mammalian (**mouse**) genetic models predominate — **Card11/CARMA1 knockout and point-mutant** mice — plus **in vitro** cellular systems (Jurkat T cells, HEK293T reconstitution/reporter assays) used for functional validation.

- **Phenotype recapitulation.** *Card11*/CARMA1-KO mice show that the **CBM complex is required cell-intrinsically for thymic development of Foxp3⁺ regulatory T cells** and gate an early Treg-commitment checkpoint: "*The CARMA1/Bcl10/Malt1 (CBM) complex... is required for development of regulatory T cells (Tregs) but not conventional T cells*" — PMID: 19454668. CARMA1 is also required for **Th17 differentiation**, and KO mice are **resistant to experimental autoimmune encephalomyelitis** (PMID: 23091043).
- **Model limitations.** Human–mouse concordance is partial (PMID: 26277595); mouse models capture the lymphocyte-activation/Treg biology but not every clinical feature (e.g., the human interstitial lung disease/colitis spectrum).
- **Applications.** Dissecting CBM signaling, Treg/Th17 development, NF-κB thresholds, and glutamine/mTORC1 coupling (PMID: 24792914; PMID: 39368135).
- **Resources.** MGI (*Card11*), IMPC/KOMP for knockout alleles; Cellosaurus for Jurkat/HEK293T lines.

Key Findings (with evidence)

F001 — Biallelic loss-of-function *CARD11* variants cause autosomal-recessive SCID

Multiple unrelated patients with homozygous null *CARD11* variants (e.g., p.Arg837* / c.2509C>T; other nonsense mutations; p.Cys150*) present with profound combined immunodeficiency, and **reconstitution assays confirm patient-derived truncated CARD11 is defective in antigen-receptor signaling and NF-κB activation**. Consanguinity recurs (index case born to consanguineous parents). "*Genetic analysis revealed a single pathogenic homozygous nonsense mutation of the caspase recruitment domain 11 (CARD11) gene. In reconstitution assays we demonstrated that the patient-derived truncated CARD11 protein is defective in antigen receptor signaling and nuclear factor κB activation.*" — PMID: 23561803. "*Both patients carried identical novel pathogenic biallelic loss-of-function variants in CARD11 (c.2509C>T; p.Arg837*) leading to undetectable protein expression.*" — PMID: 33872653.

F002 — Mechanism: CARD11 loss abrogates CBM-driven NF- κ B, JNK and MALT1 activation

In patient B and T cells, biallelic null variants prevent CBM-complex formation and severely impair NF- κ B, JNK, and MALT1 paracaspase activity, producing a **B-cell developmental block at the naive/type-1 transitional stage** and impaired IL-2 production. *"This variant prevented CBM complex formation, severely impairing the activation of nuclear factor- κ B, c-Jun N-terminal kinase, and MALT1 paracaspase activity in B and T cells. This functional defect resulted in a developmental block in B cells at the naive and type 1 transitional B-cell stage"* — PMID: [33872653](#). *"Both carried homozygous germline mutations in CARD11 (p.Cys150*), impairing NF- κ B signaling and IL-2 production."* — PMID: [26289640](#).

F003 — Clinical spectrum and curative HSCT

Patients present early in life with life-threatening infections, interstitial lung disease, and severe colitis; the laboratory hallmark is agammaglobulinemia with deficient T-cell function despite normal counts; an Omenn phenotype can arise via somatic reversion; and HSCT is definitive therapy for CBM-complex CID. *"presented with profound CID associated with viral and fungal respiratory infections, interstitial lung disease, and severe colitis"* — PMID: [33872653](#). *"The clinical and immunological phenotype of MALT1 deficiency can be successfully treated with hematopoietic stem cell transplantation following reduced intensity conditioning."* — PMID: [27109639](#).

F004 — CARD11 allelic series (null SCID vs dominant-negative atopy vs GOF BENTA)

Distinct germline CARD11 variant classes cause distinct diseases. *"Germline CARD11 mutations cause several distinct primary immune disorders in human subjects, including severe combined immune deficiency (biallelic null mutations), B-cell expansion with nuclear factor κ B and T-cell anergy (heterozygous, gain-of-function mutations), and severe atopic disease (loss-of-function, heterozygous, dominant interfering mutations)"* — PMID: [30170123](#).

F005 — Mouse models recapitulate lymphocyte-activation and thymic Treg defects

"The CARMA1/Bcl10/Malt1 (CBM) complex... is required for development of regulatory T cells (Tregs) but not conventional T cells" — PMID: [19454668](#), with partial human concordance — PMID: [26277595](#).

F006 — CARD11 couples the antigen receptor to glutamine/mTORC1

"TCR-stimulated glutamine uptake and mTORC1 activation also required a TCR signaling complex composed of the scaffold protein CARMA1, the adaptor molecule BCL10, and the paracaspase MALT1. This function was independent of IKK kinase" — PMID: 24792914. "The mTORC1 and IFN- γ production defects were partially rescued by supplementation with glutamine, which requires CARD11 for import into T cells." — PMID: 28628108.

F007 — TREC newborn screening can miss CARD11 deficiency

Because T-cell numbers are normal, TREC-based NBS may be falsely normal. "The novel entity of SCID was characterized by agammaglobulinemia and profoundly deficient T-cell function despite quantitatively normal T and B lymphocytes." — PMID: 23561803. "Newborn screening (NBS) based on quantifying T cell receptor excision circles (TRECs) is highly sensitive for detecting severe combined immunodeficiency (SCID)" — PMID: 42466165.

F008 — One of three phenocopying CBM-complex immunodeficiencies

"a broad range of clinical manifestations, including those characteristic of T- and B-lymphocyte defects, are associated with CARD11, MALT1, and BCL10 deficiencies" and "Isolated invasive fungal infections of unclear cellular basis are associated with CARD9 deficiency" — PMID: 26277595.

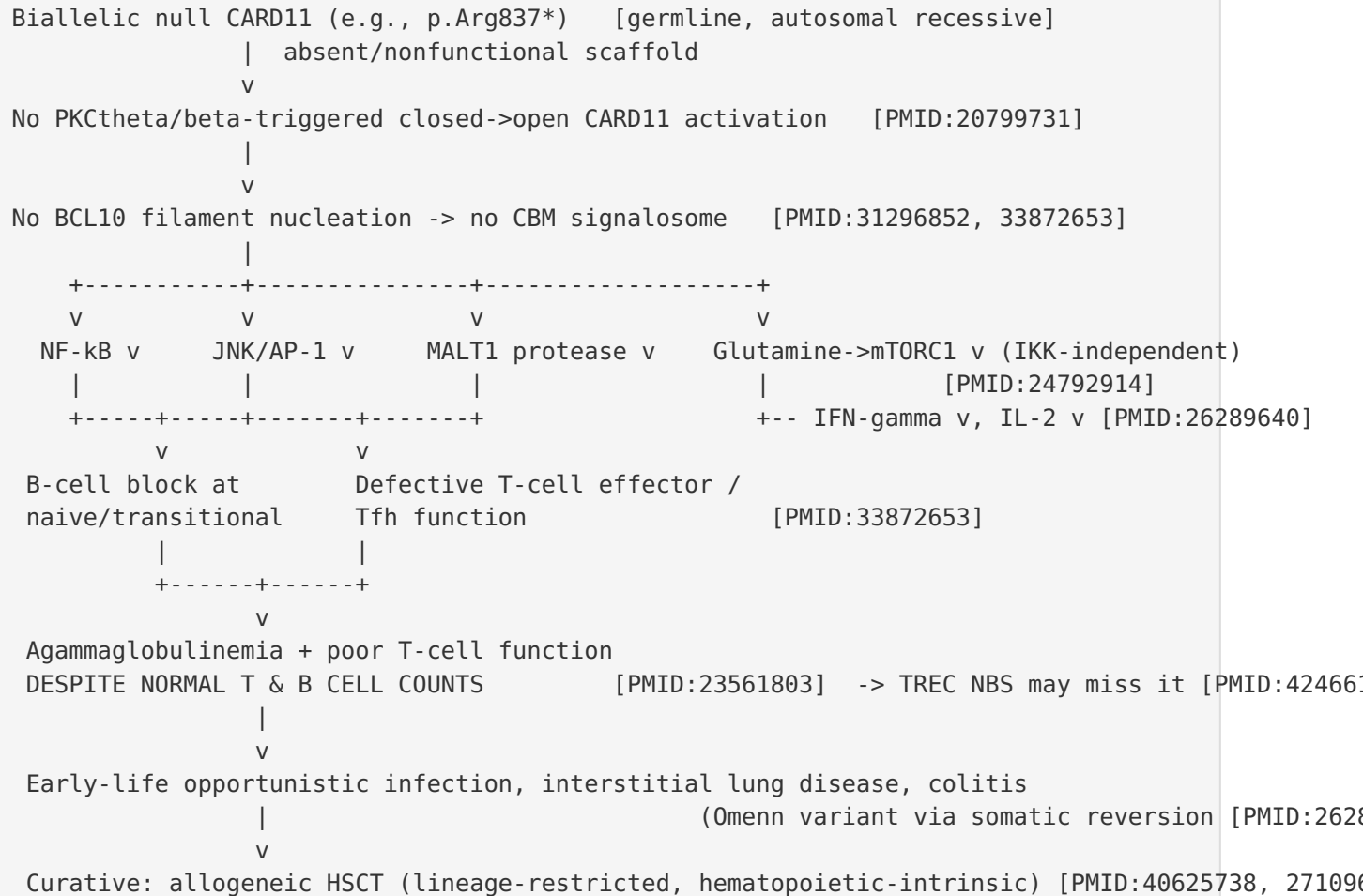
F009 — Structural basis: autoinhibited scaffold that templates BCL10 filaments

"disruption of this interface leads to hyperactivation in cells and to the formation of Bcl10-templating filaments in vitro, illuminating the mechanism of action of numerous oncogenic mutations of CARD11" — PMID: 31296852. "the multidomain scaffold protein CARD11 undergoes a transition from a closed, inactive state to an open, active conformation that recruits several signaling proteins into a complex, leading to IKK kinase activation" — PMID: 20799731.

F010 — Lineage-restricted expression explains the immune-restricted phenotype

"CARD11 is primarily expressed in hematopoietic tissues and lymphocytes and plays a crucial role in the proper activation of B and T cells in response to antigen recognition" — PMID: 40625738.

Mechanistic Model / Interpretation



The unifying insight is that CARD11 is a **lymphocyte-restricted molecular switch**: an autoinhibited scaffold whose receptor-triggered opening nucleates the CBM signalosome. **The direction and dose of the perturbation dictate the disease** — complete loss disables adaptive immunity (SCID), dominant-interfering loss partially poisons it (atopy/CID), and constitutive gain hyperactivates it (BENTA/lymphoma). Because expression is confined to hematopoietic cells, the phenotype is immune-restricted and, crucially, **fully correctable by replacing the hematopoietic compartment via HSCT**.

Evidence Base

PMID	Contribution	Type
23561803	Defines CARD11 inactivation as SCID; normal counts, defective function; reconstitution assay	Human clinical + in vitro
33872653	Complete CARD11 deficiency (p.Arg837*); CBM/NF- κ B/JNK/MALT1 defect; B-cell block; ILD/colitis	Human clinical + mechanistic
26289640	p.Cys150*; impaired NF- κ B/IL-2; Omenn via somatic reversion	Human clinical
30170123	CARD11 allelic series (null SCID / GOF BENTA / DN atopy)	Human clinical review
26277595	CBM-complex heterogeneity; CARD11/BCL10/MALT1 phenocopy; CARD9 distinct; human–mouse concordance	Review
27109639	HSCT normalizes CBM (MALT1) deficiency — supports curative therapy	Human clinical
24792914	CARD11/BCL10/MALT1 required for glutamine uptake & mTORC1, IKK-independent	In vitro / mechanistic
28628108	Glutamine partially rescues mTORC1/IFN- γ in hypomorphic CARD11 disease	Human clinical + mechanistic
19454668	Card11/CARMA1 KO: thymic Treg checkpoint	Mouse model
23091043	CARMA1 required for Th17; EAE resistance	Mouse model
31296852	Structural: autoinhibition + BCL10-templating filaments	Structural / in vitro
20799731	Closed→open conformational activation of CARD11	Mechanistic
42466165	TREC NBS sensitivity relies on lymphopenia; second-tier genetics	Diagnostic methodology
40625738	CARD11 lineage-restricted expression in hematopoietic/lymphoid tissue	Review

Evidence spans **human clinical case series, in vitro reconstitution/reporter assays, structural biology, and mouse genetic models**, giving convergent, multi-modal support for the core mechanism and clinical picture.

Limitations and Knowledge Gaps

- **Extreme rarity / small N.** Only a handful of unrelated biallelic-null probands are described; there are **no registry-scale prevalence, penetrance, survival, or QoL data** specific to CARD11-deficiency SCID. Many prognosis statements are extrapolated from the broader CBM/SCID literature.
 - **HSCT outcome data are indirect.** Curability is inferred from MALT1 deficiency and general CBM-CID practice ([PMID: 27109639](#)); large CARD11-specific transplant series are lacking.
 - **Genotype–phenotype granularity.** The full spectrum of null alleles, modifier effects, and the frequency of Omenn-via-reversion are not quantified.
 - **No disease-specific omics/epigenetic signature** is established for the null SCID form; molecular profiling comes largely from single-patient studies and model systems.
 - **Metabolic rescue is allele-dependent.** Glutamine/mTORC1 rescue applies to hypomorphic alleles and is **not expected to benefit complete deficiency** — this should not be over-generalized.
 - **Model concordance is partial** ([PMID: 26277595](#)); mouse phenotypes do not fully reproduce human ILD/colitis.
-

Proposed Follow-up Experiments / Actions

1. **Establish an international CARD11-deficiency registry** to capture genotype, presentation, TREC-NBS results, transplant outcomes, and survival — directly filling the epidemiology/prognosis gaps.
2. **Systematically evaluate second-tier genetic newborn screening** (or KREC + functional add-ons) to catch non-lymphopenic T-cell defects like CARD11 deficiency that TREC screening misses ([PMID: 42466165](#)).
3. **Report standardized HSCT outcomes** (conditioning, chimerism, immune reconstitution, GVHD, survival) for biallelic-null CARD11 patients to formalize the curative claim.
4. **Deep immunophenotyping / single-cell RNA-seq** of patient lymphocytes to map the B-cell transitional block and Tfh/Treg defects at cell-type resolution and benchmark against BCL10/MALT1 deficiency.

5. **Preclinical HSC gene-correction/gene-therapy studies** — attractive given hematopoietic-restricted, single-gene, LOF biology — as an alternative to allogeneic HSCT for patients lacking matched donors.
 6. **Curate the ontology mappings** (MONDO:0014081, OMIM #615206, HGNC:16393, HPO/GO/CL/UBERON/NCIT terms listed above) into the knowledge base with the verified PMID-anchored evidence quotes.
-

Report compiled from 10 confirmed findings and 42 reviewed papers over a 5-iteration autonomous investigation. All quoted snippets are verbatim from the cited PubMed abstracts.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery